

From field observation to traceable care.

Release report · 16 September 2026 UTC

Implemented: English/Telugu field workflows, mobile layouts, offline image inference, batch care and outcomes, account/farm authorization, and a tested recovery path.

Evidence boundary: This is a working application with internal image-classification evidence. Disease diagnosis, prospective warning performance and yield benefit remain unvalidated.

1. Product and reused foundation

React, TypeScript and Vite provide the UI; i18next provides bundled English and Telugu copy. Dexie/IndexedDB stores device records and media. Express, SQLite and Express sessions provide the account API. Torchvision MobileNetV3-Small supplies ImageNet-pretrained convolutional features; a supervised classification head is trained on the audited larval image source. ONNX Runtime executes the same frozen graph on the server and in browser WebAssembly.

Camera / upload

Image evidence

Batch assessment

Explainable care

Follow-up / review

Observed outcomes

Deferred sync

The product contribution is the connection between uncertain observations, image evidence, preventive care, follow-up and recorded outcomes. No new convolutional architecture is claimed. A low image score does not erase reported deaths or environmental concerns. Self-reported fresh feed is accompanied by a suspect-feed sensitivity scenario.

Capability	Implemented behavior
Language and mobile	Telugu/English selection persists; forms retain drafts during switching; Telugu font and translations are bundled for offline use. Mobile navigation, forms, dialogs, tables and explanations adapt to narrow screens.
Capture and analysis	Camera, uploads and sampled video; quality checks and occlusion explanations. Explicit server/device inference selection with model checksum.
Records and care	Append-only assessments, interventions, reviews, follow-ups, observed onset and harvest outcomes. Missing measurements remain unknown.
Supervision	Farm, risk and latest-assessment date filters; visit priorities, descriptive trends and outcome comparisons.
Security	Hashed passwords, persisted sessions, login throttling, farm-scoped sync, supervisor-only review writes, linked-record checks and authenticated submission receipts. Separate local namespaces per account/assignment.
Deployment	Validated Caddy HTTPS configuration, account service template, verified SQLite backup and recovery commands. Public DNS and host activation pending.

2. Internal model evidence

The larval source is [Mungase and Chiwhane, Silkworm Diseases Dataset v1 \(CC BY 4.0\)](#). The acquisition contains 5,529 images. The audit retained 3,843 source-family representatives and quarantined conflicting groups. Training, selection, calibration and test partitions are separate.

Measure	Internal result	Interpretation
Test accuracy	98.83%	Single-source image holdout only
Macro F1	0.9835	Appearance classes, including Overlap
Network size	3,732,850 bytes	Complete exported MobileNetV3 graph

The backbone remains frozen; this is transfer learning of a classification head. Overlap is a scene condition, not a disease. The separate mulberry leaf classifier provides experimental feed-image review and cannot establish larval health. The overall

low/medium/high priority score uses transparent engineering rules, not a trained predictor of future mortality.

- Dataset folder labels are not independently confirmed diagnoses.
- No linked onset, environmental or mortality data: no early-warning lead time or yield benefit established.
- One source, heuristic duplicate groups; hidden shared specimens/capture sessions and background confounding remain possible.
- Overlap is not a disease; its output is withheld from health interpretation.
- Temperature calibration is internal to this source; not validated field confidence.
- Unsupported objects may still receive high scores. No universal out-of-distribution detector.
- Random image-group holdout is not an independent farm or regional validation.

3. Verification and evaluation pathway

See the repository testing record for exact suite counts and commands. Browser tests exercise backend/device probability parity, camera permission handling, offline inference, capture/reload/sync, accessible screens, Telugu workflows and mobile widths. Account tests exercise anonymous rejection, cross-farm isolation, review permissions, logout, offline expiry and user workspace separation. Python evaluator tests check leakage rejection and parity with the frozen graph. These are software checks, not field-effectiveness studies.

The external evaluator consumes an independently supplied manifest with farm, batch, capture group, source, licence and label method. It rejects empty data and exact overlap with the acquisition corpus and reports confusion matrices, per-farm results, calibration error, score abstention and farm-cluster uncertainty. It does not tune the network or thresholds. Exact hashes cannot establish full acquisition independence; near-duplicate and annotation review remain necessary.

Prospective warning validation needs independently timed onset, a predeclared prediction horizon, negative follow-up cohorts and a frozen model. Yield-benefit claims need a comparison design that controls baseline differences and complete outcome denominators. Logging an intervention and a later harvest is insufficient to infer benefit.

4. Remaining external requirements

Requirement	Current boundary
departmentalDataset	Official link unavailable at last access check; no verified download
independentFarmAccuracy	Not established: independent labelled data not supplied
earlyWarningLeadTime	Not established: independently timed onset cohort not supplied
yieldImprovement	Not established: outcome impact study not supplied
reviewedTeluguCare	Software translation implemented; independent language/SOP review not obtained
publicDeployment	Configuration prepared; domain and host access required

Physical-phone and real-shed testing were explicitly excluded from this implementation request. Browser emulation is reported as browser emulation. Telugu copy is implemented; independent linguistic and departmental review is still required.

5. Curated source inventory

This inventory records inspected evidence and access limitations. It is a targeted review, not an exhaustive literature survey. Historical source-reported claims are not adopted as application performance.

S01 • Technology Descriptor

CSR&TI Berhampore • Undated • Official SOP

Task: Regional stage guidance

Evidence: Full PDF reviewed

Data access: Not applicable

Limit: Local breed and Andhra Pradesh applicability unverified

https://csrtiber.res.in/Technologies_descriptor.pdf

S02 • Muga Disease Early Warning System

NESAC and CMERTI • 2020 site copyright • Government project

Task: Environmental history and infestation

Evidence: Official project page reviewed

Data access: Not acquired

Limit: Different species and outdoor setting; independent prospective validation unverified

<https://apps.nesdr.gov.in/MDEWS/index.php>

S03 · Automated Disease Detection in Silkworms Using Machine Learning Techniques

Binson and Manju · 2024 · Paper

Task: Larval image classification

Evidence: Methods reviewed

Data access: Archive not located

Limit: Single-day controlled-background images; disease images from fifth day after infection

<https://journal2.upgris.ac.id/index.php/asset/article/download/965/387>

S04 · Silkworm_Diseases_Dataset v1

Kajal Mungase and Shwetambari Chiwhane · 2026-09-06 · Dataset

Task: Larval image categories

Evidence: Publisher metadata reviewed

Data access: 5529 images acquired and audited; 3843 representative groups; trained MobileNetV3 head

Limit: Overlap is a scene category; source-family independence unknown

<https://data.mendeley.com/datasets/g4b89vpp9c/1>

S05 · Silkworm Diseases v1

Silkworm Annotation · 2024-11 · Dataset

Task: Larval detection

Evidence: Publisher version page reviewed

Data access: Download interface not resolved

Limit: Augmented version images and object crops are not independent specimens

<https://universe.roboflow.com/silkworm-annotation-y6ztu/silkworm-diseases-y0bro/dataset/1>

S06 · SilkLDP implementation

hokhoi02new · 2025 · Repository

Task: Larval segmentation

Evidence: Owner README reviewed

Data access: Sample only; full data described as proprietary

Limit: MIT code license does not grant rights to unavailable full data

https://github.com/hokhoi02new/silkworm_disease_segmentation_using_deeplearning_and_computer_vision_technology

S07 · silkworm-fpga-ml

yoctocandela · 2026 · Repository

Task: Classical crop classifier

Evidence: Owner methods and data README reviewed

Data access: Images not redistributed

Limit: FPGA implementation absent; exporter split alone does not exclude leakage

<https://github.com/yoctocandela/silkworm-fpga-ml>

S08 · Smart Sericulture System

ZHAGARAM creators · 2024-10 · Hackathon project

Task: IoT and visual advisory application

Evidence: Indexed creator page and gallery

Data access: Not acquired

Limit: Pilot and multilingual/predictive capabilities described as future work

<https://devpost.com/software/zhagaram-smart-sericulture-system>

S09 · Silk Shield

Salamath et al. · 2025-05 · Paper/project

Task: Image classification and climate suggestions

Evidence: Full five-page PDF reviewed

Data access: No archive supplied

Limit: No auditable dataset count or numerical evaluation table

<https://iarjset.com/wp-content/uploads/2025/05/IARJSET.2025.125345.pdf>

S10 · WormSORT

Shi et al. · 2026-06-15 · Paper

Task: Individual video tracking

Evidence: Primary full text reviewed

Data access: OSF retrieval failed

Limit: Two sequences at one institute; tracking is not disease lead-time validation

<https://doi.org/10.1371/journal.pcbi.1014410>

S11 · Mulberry leaf dataset

Nahiduzzaman et al. · 2023 · Dataset/paper

Task: Leaf appearance classification

Evidence: Publisher card and linked paper reviewed

Data access: 1091 images acquired with original hashes

Limit: Single source; plant/session identity unknown; no larval labels

<https://www.kaggle.com/datasets/nahiduzzaman13/mulberry-leaf-dataset/data>

S12 · Mulberry cultivar dataset v3

Mendeley contributors · Version 3 · Dataset

Task: Cultivar classification

Evidence: Publisher metadata reviewed

Data access: Not acquired

Limit: Cultivar labels cannot become disease labels

<https://data.mendeley.com/datasets/ds45yy9jrc/3>

S13 · Andhra Pradesh silkworm dataset announcement

IndiaAI · Date unverified · Government data announcement

Task: Advertised images/environment/outcomes

Evidence: Official-account announcement

Data access: Official redirect resolved; AIKosh target currently unavailable; corpus not acquired

Limit: No verified sample size license or temporal linkage

https://www.linkedin.com/posts/indiaai_sericulture-opendata-aikosh-activity-7449837970519404544-cvI8

S14 · SilkWorm

Silkwormincubator · Date unverified · Repository

Task: Environmental sensor capture

Evidence: Owner repository reviewed

Data access: No disease corpus established

Limit: Instrumentation comparator; not trained early-warning evidence

<https://github.com/Silkwormincubator/SilkWorm>

S15 · E-Nose-Based VOC Analysis for Rapid Detection of Grasserie Disease

Publisher metadata · 2026 · Paper lead

Task: VOC sensors and ML

Evidence: Indexed abstract; direct 403

Data access: Not acquired

Limit: Not RGB imagery; sampling and lead time unverified

<https://www.sciencedirect.com/science/article/pii/S2590123026033839>

S16 · Rapid BmNPV detection using FTIR spectroscopy and chemometric modelling

Publisher metadata · 2026 · Paper lead

Task: Blood spectroscopy

Evidence: Indexed abstract; direct 403

Data access: Not acquired

Limit: Requires a different sensing modality; performance not adopted

<https://www.sciencedirect.com/science/article/pii/S1386142526001708>
